Animesh Kumar

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LinkedIn
GitHub
Portfolio

Professional Summary

Artificial Intelligence Engineer (MS, GPA 4.0) with experience building end-to-end recommendation, retrieval, and agentic AI systems. Skilled in learning-to-rank models, transformer-based architectures, and production AI deployment using PyTorch, Spark, and AWS. Published author in deep learning and computer vision (ACCAI 2024, INFUS 2024), focused on optimizing user engagement and personalization through data-driven ML solutions.

Education

Arizona State University, MS in Computer Science – Tempe, AZ — May 2026

- GPA: 4.0/4.0

MIT World Peace University, BTech in ECE – Pune, India — July 2024

- GPA: 3.9/4.0

Technical Skills

Languages & Frameworks: Python, MongoDB, PyTorch, TensorFlow, Keras, HuggingFace Transformers, XGBoost, scikit-learn, SHAP

ML & AI: Recommender Systems, NLP, Computer Vision, CNNs, RNNs, Transformers, LLMs, RAG, Agentic AI, LangGraph, CrewAI, AutoGen, Information Retrieval, Model Evaluation, Prompt Engineering, Vector Databases

Data & MLOps: Apache Spark, Pandas, NumPy, Neo4j, FastAPI, Docker, Kubernetes, AWS, GCP, Cloud Run, Vertex AI, A/B Testing, CI/CD, Git, NoSQL, Distributed Systems, Model Monitoring, Unit Testing

Experience

IT Specialist / Data Scientist, Trinity Tree – Remote — June 2026 – Present

- Co-built a production LangGraph pipeline on Cloud Run, rebuilding a local-only no-code prototype to draft posts for 3 platforms via a cross-provider writer–critic loop (GPT-5.5, Gemini) that checks claims against the practice's website
- Owned the approval layer (durable pause/resume, 24h reminder, 48h expiry) and fixed a flaw where email security scanners silently executed review decisions, by splitting each decision into a read-only confirmation page and a POST action

Software Engineering Intern, SAMSAN Technologies – Pune, India — July – December 2023

- Developed a CNN-based smoking detection model (VGG16, transfer learning) achieving 88% accuracy at 15 FPS, then integrated it into a real-time driver monitoring pipeline with facial landmark-based drowsiness detection, applying probability thresholding to minimize false alarms

Projects

MatchDay: Agentic AI Assistant with MCP Integration and RAG Pipeline — Summer 2026

- Built conversational football agent orchestrating Gemini 3.1 Pro through a custom Python tool-calling loop with 21 tools spanning grounded web search, MongoDB MCP for user state, and Discovery Engine RAG; deployed on Cloud Run
- Raised RAG top-2 hit rate from 58% to 92% by building a 28-query labeled eval harness and diagnosing a relevance threshold that discarded correct results

Ticket Router: Fine-Tuned LLM Pipeline for Support Triage — Spring 2026

- Fine-tuned DistilBERT classifier and Llama 3.2 3B with QLoRA (4-bit, r=16) on ASU's A100 supercomputer, achieving 99.9% accuracy on 4K held-out samples; chained into an end-to-end inference pipeline deployed on HuggingFace Spaces with a Gradio frontend

Recon: Automated Job Search Pipeline with Local LLM — Spring 2026

- Built automated pipeline scraping 3 sources (LinkedIn, Greenhouse, Lever APIs) across 18 companies daily, screening jobs via local LLM (Qwen 2.5 7B) with fit classification filtering 40% of listings as mismatched
- Implemented LaTeX-based resume tailoring with section-anchored rewriting, cold outreach drafting, and bidirectional Google Sheets sync for human-in-the-loop approval; runs on $0/month using Apify free tier and local GPU inference

Production E-commerce Recommendation System — Summer 2025

- Built hybrid recommendation engine processing 185K+ interactions across 10K users achieving 25% conversion rate (2x collaborative filtering, 3x content-based) with sub-50ms P99 latency; deployed on AWS ECS with Docker and FastAPI
- Designed A/B testing framework with Chi-square and ANOVA analysis across 3 model variants to measure ranking quality and recommendation relevance